

Chapter 14.11

ResNet50 Theory & Scripting

தமிழில்

Less than 30 Mins

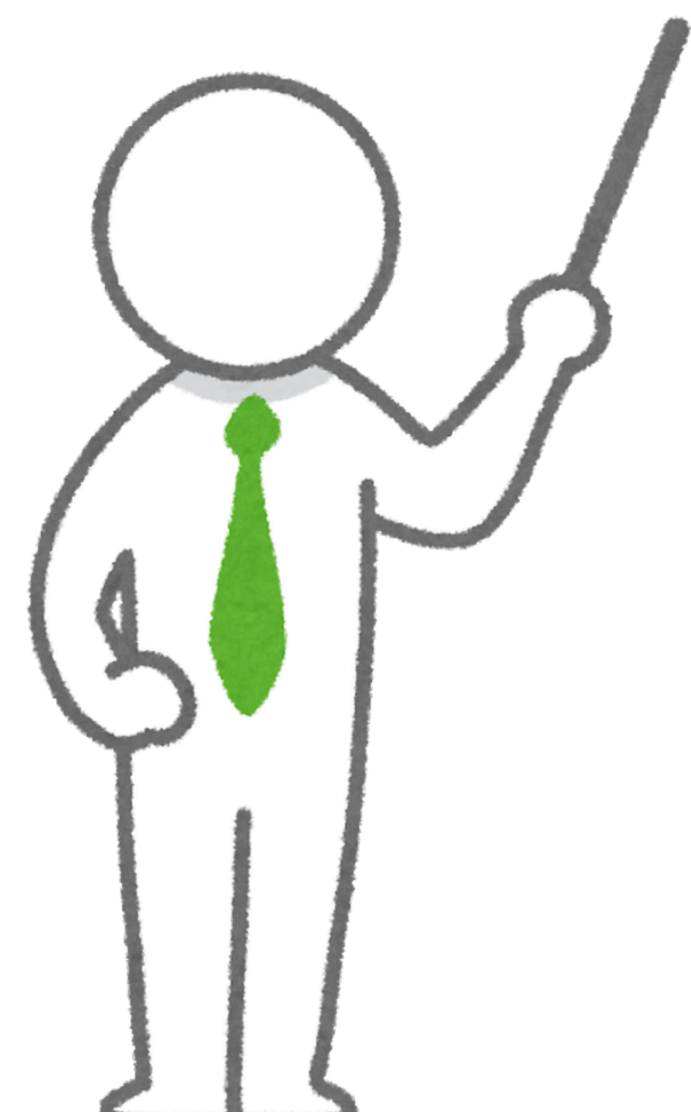
AI COACH JOHN

CEO & FOUNDER

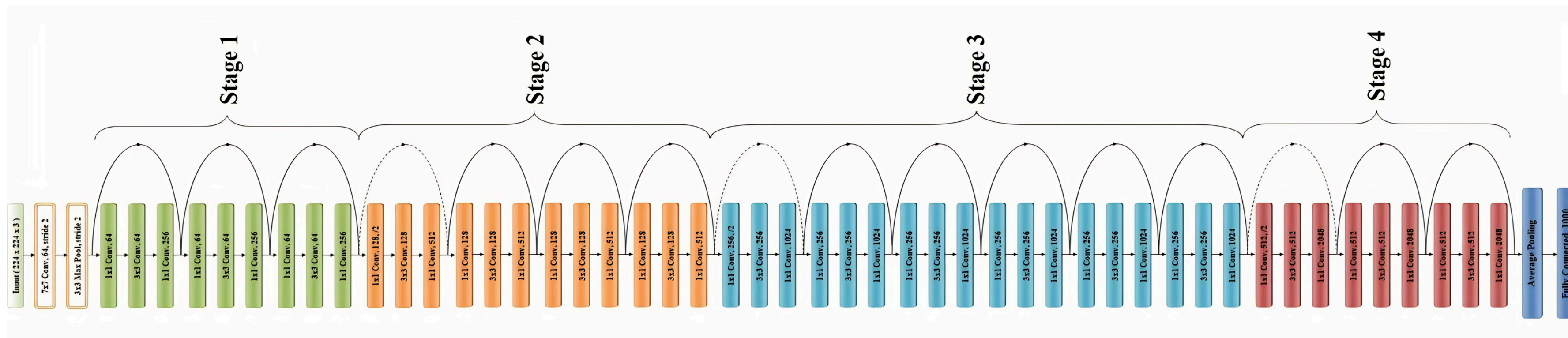
~~**Vanishing Gradient**~~



Permanently Solved



ResNet 50 Architecture



*Understanding this, opens 2 More
architectural Understanding*

ResNet-101

ResNet-152

Your Interview Key Pointers (*Know the Principle*)

A Complete architectural rethink

☞ **1. The Problem** = *Vanishing gradients*

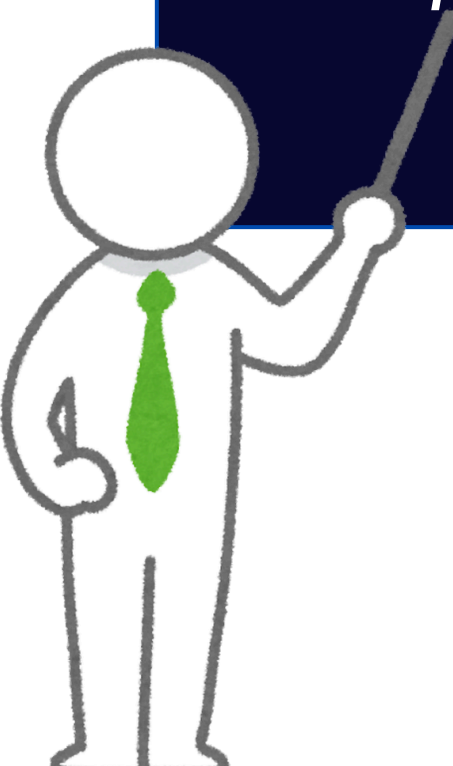
☞ **2. The Skip Connection** = $y = F(x) + x \rightarrow$ let layers learn the residual, not full output.

☞ **1. Identity Block**

When **input shape == output shape** · skip path is just a wire.

☞ **2. Convolution Block**

When **input shape $\neq$ output shape** · skip path needs 1×1 conv to reshape.



Key Takeaway Point

- **LeNet** → proved *machines can see...*
- **AlexNet** → proved *depth works*
- **VGG** → proved **Simplicity + depth** *works even better.*
- **GoogLenet(Inception V1)** → proved **Depth + Parallel** *is more efficient*
- **Inception V2** → proved Deep learning **Stablity** *matters*
- **Inception V3** → proved Deep learning **Efficiency** *matters*
- **ResNet 50** → proved Deep Networks must *learn residuals to scale*

Scripting????

